



Selected Topics on Complex Algebraic Varieties - MPG3320 III
Assignment 1
August 31, 2026

Problem 1. Since the ideal (x, y) is maximal, it suffices to show that

$$I = (x + xy, y + xy, x^2, y^2) \supseteq (x, y)$$

Indeed, we observe the following:

$$(x + xy)(y + xy) = xy + x^2y + xy^2 + x^2y^2$$

Since $x^2, y^2 \in I$, it follows that $xy \in I$, which implies that $x, y \in I$, and therefore $(x, y) \subseteq I$.

Problem 2.

- (a) The set S is not an affine variety; otherwise, it would be a Zariski-closed subset and hence a closed under the Euclidean topology, which is a contradiction.
- (b) Let us consider the ideal

$$I = (x^{p-1} - 1, x - y) \subseteq \mathbb{F}_q[x, y]$$

Given $(a, a) \in S$, it follows from Fermat's Little Theorem that

$$a^{p-1} - 1 = 0 \quad \text{and} \quad a - a = 0$$

then $(a, a) \in V(I)$. On the other hand, if $(x, y) \in V(I)$, we see that $x = y$ and $x^{p-1} = 1$, that is, $x \neq 0$ which implies that $(x, y) \in S$.

- (c) We define

$$\mathcal{U} := U_0 \cap U_1 \subseteq \mathbb{P}_{\mathbb{C}}^1 \quad \text{and} \quad \mathcal{V} = \overline{V(xy - 1)} \cap U_2 \subseteq \mathbb{P}_{\mathbb{C}}^2$$

where $U_i = \{[x_0 : \cdots : x_n] \in \mathbb{P}_{\mathbb{C}}^n : x_i \neq 0\}$. Since the ideal $(xy - 1)$ is generated by a single element, its homogenization is $(xy - z^2)$, and thus $\mathcal{V} = V^p(xy - z^2) \subseteq U_i$ for $i = 0, 1, 2$.

We note that both $\mathcal{U}$ and $\mathcal{V}$ are quasi-projective varieties. Furthermore, through the natural identification, $\mathbb{C}^*$ and $V(xy - 1)$ are homeomorphic to $\mathcal{U}$ and $\mathcal{V}$, respectively. On the other hand, the map

$$z \mapsto (z, z^{-1})$$

is a correspondence between the points in $\mathbb{C}^*$ and those in $\mathcal{V}(xy - 1)$, but it is not a polynomial application. However, given $[x : y] \in \mathcal{U}$ and under these identifications, we have the following:

$$[x : y] = [xy^{-1} : 1] \mapsto xy^{-1} \mapsto (xy^{-1}, x^{-1}y) \mapsto [xy^{-1} : x^{-1}y : 1] = [x^2 : y^2 : xy]$$

The map $\varphi : \mathcal{U} \rightarrow \mathcal{V}$ given by $[x : y] \mapsto [x^2 : y^2 : xy]$ is a morphism of quasi-projective varieties, because it is polynomial on each coordinate. Moreover, it is an isomorphism with inverse $\psi([x : y : z]) = [x : z]$. We observe the following:

$$\psi\varphi([x : y]) = \psi([x^2 : y^2 : xy]) = [x^2 : xy] = [x : y]$$

and

$$\varphi\psi([x : y : z]) = \varphi([x : z]) = [x^2 : z^2 : xz] = [x^2 : xy : xz] = [x : y : z]$$

Which proves that they are isomorphic as quasi-projective varieties.

Problem 3.

- (a) First, we will prove the following lemma:

Lemma: Let $f \in \mathbb{C}[x_1, \dots, x_n]$ be non-zero. Then $f^{-1}(0)$ has empty interior.

Proof. We will proceed by induction on $n \in \mathbb{N}$. If $n = 1$, we know that f has finitely many roots. If $f^{-1}(0)$ had a non-empty interior, it would contain infinitely many roots, yielding a contradiction. For the inductive step, suppose, by contradiction, that there exists an open set $U \subseteq \mathbb{C}^n$ such that $U \subseteq f^{-1}(0)$. We write:

$$f = \sum_k f_k(x_1, \dots, x_{n-1})x_n^k \quad \text{with } f_k \in \mathbb{C}[x_1, \dots, x_{n-1}]$$

Let $a = (a_1, \dots, a_n) \in U$, and consider the continuous function

$$\begin{aligned} i : \mathbb{C} &\rightarrow \mathbb{C}^n \\ z &\mapsto (a_1, \dots, a_{n-1}, z) \end{aligned}$$

We observe that $i^{-1}(U)$ is a non-empty open set in $\mathbb{C}$ and $g(z) = f(a_1, \dots, a_{n-1}, z)$ is a polynomial that vanishes on an open set of $\mathbb{C}$. By the base case, it follows that $g \equiv 0$. Therefore, $f_k(a_1, \dots, a_{n-1}) = 0$ for all k .

The previous argument holds for all $a \in U$. Since the map $a \mapsto (a_1, \dots, a_{n-1})$ is open, the polynomial f_k vanishes on an open set, and by the inductive hypothesis, $f_k \equiv 0$ for all k . Thus, we conclude that $f^{-1}(0)$ has an empty interior. $\square$

The lemma immediately implies that given a non-constant $f \in \mathbb{C}[x_1, \dots, x_n]$, the set $f^{-1}(\mathbb{C} \setminus \{0\})$ is dense in $\mathbb{C}^n$.

Let V be a proper affine variety. Then $V = V(I)$ for some non-zero ideal $I \subseteq \mathbb{C}[x_1, \dots, x_n]$. By the Hilbert's Basis Theorem, there exist polynomials $f_1, \dots, f_m$ such that $I = (f_1, \dots, f_m)$. Thus

$$V = V(I) = V(f_1, \dots, f_m) = \bigcap_{i=1}^m V(f_i) \subseteq V(f_1)$$

By taking complement and its closure, we obtain that V^c is dense in $\mathbb{C}^n$. That is, every Zariski-open set is dense under the Euclidean topology.

- (b) Suppose, by contradiction, that both topologies are identical. Consider the Zariski-closed subset $V = V(x - y) \subseteq \mathbb{A}_{\mathbb{C}}^2$. Therefore, V is closed under the product topology.

The previous statement is equivalent to saying that the Zariski topology in $\mathbb{A}_{\mathbb{C}}^1$ is Hausdorff. However, it is well known that this topology is identical to the cofinite topology on $\mathbb{C}$ (since every ideal is generated by a single element, and thus every affine variety has finitely many elements), which is non-Hausdorff on infinite sets. This contradiction proves the assertion.

Problem 4.

- (a) We see that the ideal $I := (x_1x_2) \subseteq k[x_1, \dots, x_n]$ is not prime. Indeed, note that $x_1x_2 \in I$ but $x_i \notin I$. However, it remains true that I is a radical ideal.

If $f^n \in I$, by definition x_1x_2 divides f^n . In particular, $x_i \mid f^n$, and since each x_i is prime, it follows that $x_i \mid f$. Because x_1 and x_2 are coprime, we have that $x_1x_2 \mid f$, which concludes the proof.

- (b) $\Rightarrow$ | Suppose that $I(V)$ is not prime. Then, there exist polynomials f_1, f_2 such that $f_1f_2 \in I(V)$ but $f_i \notin I(V)$. We observe that

$$V = (V \cap V(f_1)) \cup (V \cap V(f_2))$$

Additionally, $(V \cap V(f_i)) \neq V$ since there exist points $p_i \in V$ such that $f_i(p_i) \neq 0$. Consequently, V is reducible.

- $\Leftarrow$ | Suppose that V is reducible, then

$$V = V_1 \cup V_2$$

where each $V_i \neq V$ is a proper affine variety. There exist points $p_i \in V \setminus V_i$ and polynomials f_i such that $f_i|_{V_i} \equiv 0$ and $f_i(p_i) \neq 0$. We observe that $f_1f_2 \in I(V)$ but $f_i \notin I(V)$, which completes the proof.

Problem 5. Let us see the following lemma:

Lemma: Let k be a field that is not algebraically closed. Then, there exists a polynomial $q \in k[x, y]$ such that $q(x, y) = 0$ if and only if $x = y = 0$.

Proof. Since k is not algebraically closed, there exists a non-constant polynomial $f \in k[x]$ without roots in k . We write

$$f = a_0 + a_1x + \cdots + a_nx^n \quad \text{with } a_0, a_n \neq 0$$

We define $q(x, y) \in k[x, y]$ by

$$q(x, y) := \tilde{f} = a_0y^n + a_1xy^{n-1} + \cdots + a_{n-1}x^{n-1}y + a_nx^n$$

Let $(x, y) \in k^2$ be such that $q(x, y) = 0$ and suppose by contradiction that $y \neq 0$. Otherwise, we would have $a_nx^n = 0$, which implies that $x = 0$. Therefore,

$$0 = a_0 + a_1\left(\frac{x}{y}\right) + \cdots + a_n\left(\frac{x}{y}\right)^n = f\left(\frac{x}{y}\right)$$

However, the above equation has no solutions in k , by the choice of f . We conclude that $q(x, y) = 0$ if and only if $x = y = 0$. $\square$

Let $V = V(f_1, \dots, f_m) \subseteq \mathbb{A}_k^n$ be an affine variety and let $q \in k[x, y]$ be as above, we define the following polynomials recursively:

$$q_i := q(f_i, q_{i+1}) \quad \text{for } 1 \leq i < m$$

and $q_m = f_m$. We claim that $V(q_1) = V$. Indeed, we have that $q_1 = 0$ if and only if $f_1 = q_2 = 0$. Similarly, $q_2 = 0$ is equivalent to saying that $f_2 = q_3 = 0$. Inductively, it follows that $q_1 = 0$ if and only if $f_i = 0$ for all i , as desired.

Problem 6.

- (a) Let $p \in C(Y)$ be non-zero. By definition $[p] \in Y$, which implies that $f(p) = 0$ for all homogeneous polynomials $f \in I$. Let $f \in I$, and write

$$f = \sum_r f_r \quad \text{with } f_r \text{ homogeneous}$$

Since the ideal I is homogeneous, $f_r \in I$ for all r . Thus, $f_r(p) = 0$ and consequently $f(p) = 0$, it follows that $p \in V(I)$.

Let $p \in V(I)$ be non-zero; therefore, $f(p) = 0$ for all $f \in I$. In particular, this holds homogeneous polynomial $f \in I$, which implies that $[p] \in Y$. That is, $p \in C(Y)$.

- (b) Similarly to the case of affine varieties, it is true that a projective variety V is irreducible if and only if $I^h(V)$ is a prime ideal. Thus, it suffices to show that $I(C(Y)) = I^h(Y)$.

Let $f \in I(C(Y))$. Then, it follows that $f|_{C(Y)} \equiv 0$. We express f as

$$f = \sum_r f_r \quad \text{with } f_r \text{ homogeneous}$$

We want to show that $f_r|_Y \equiv 0$ for all r . Given a point $[p] \in Y$, we observe that $p \in C(Y)$, which implies that $f(p) = 0$. Moreover, $f(\lambda p) = 0$ for all $\lambda \neq 0$. Consequently, the polynomial

$$g(\lambda) = f(\lambda p) = \sum_r f_r(p)\lambda^r$$

has infinitely many roots, which implies that $g \equiv 0$. Therefore, $f_r(p) = 0$ for all r . We conclude that $f \in I^h(Y)$.

Given $f \in I^h(Y)$, we write f as

$$f = \sum_r f_r \quad \text{with } f_r \text{ homogeneous and } f_r|_Y \equiv 0$$

Fix a non-zero point $p \in C(Y)$. Again, by definition, we have that $[p] \in Y$. Therefore, $f_r(p) = 0$ for all r , which implies that $f(p) = 0$. In summary, it follows that $f \in I(C(Y))$, as desired.